Vincent S. Pronel

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Objective

Computer Science undergraduate with a strong foundation in **SQL**, **Database Management**, and **full-stack development**, seeking a **software engineering or data-focused internship** to apply **query optimization**, **schema design**, and **production system architecture** to ship production web applications.

Education

Kean University — **Union, NJ**
Bachelor of Science in Computer Science — Expected May 2027
- **Relevant Courses**: Data Structures, Computer Organization & Architecture, Database Management Systems, Computer Operating Systems, Object Oriented Analysis & Design
- **Dean's List:** Spring 2024, Fall 2025

Skills

Languages: Java, Python, SQL, PHP, JavaScript, HTML/CSS (Tailwind), LaTeX
Frameworks & Libraries: React, Next.js, Node.js, Express, REST APIs, Google Charts
Database & Infrastructure: MySQL, PostgreSQL, Supabase, Schema Design (ERDs), Query Optimization, Apache/XAMPP, GitHub Actions, Vercel (CI/CD)
Tools & Other Skills: Linux (Ubuntu), Git/GitHub, VS Code, MySQL Workbench, Figma

Experience

Data Visualization & Analytics Intern — July 2026
Group Summer Scholars Research Program, Kean University — *Union, NJ*
- Led daily labs for the **Visualization and Analytics Development for Data Insights** stream under **Dr. Ching-yu Huang**, teaching **5 high school students SQL**, **PHP**, **MySQL**, **JavaScript**, and **Google Charts**; authored instructional decks and **debugging** guides progressing students from raw datasets to **dynamic database-driven** sites deployed via **Infinity Free**.

Computer Science Tutor (Code Samurai) — Jan. 2026 – Present
Kean University — *Union, NJ*
- Hand-selected by CS faculty, leading **twice-weekly one-on-one sessions** covering **Database Management Systems**, **SQL query optimization**, and **Data Structures**; guided students through complex **schema design** problems and **algorithm analysis**.

Web Developer — Jan. 2026 – Present
WiCyS (Women in Cyber Security)
- Building and maintaining the official **full-stack website** for the Kean University WiCyS chapter, shipping an **events and announcements system** and **resource hub** backed by **REST API** integrations on **production-level infrastructure**.

Computer Technician — Jan. 2022 – Aug. 2024
Old Bridge Township School District — *Middlesex County, NJ*
- Managed district-wide **software deployment** and **OS configuration**, performing **hardware repairs** on **Windows systems** across 7 schools to ensure consistent classroom environments.

Professional Development & Research

CRA UR2PhD-REU — June 2026 – Present
Undergraduate Research Fellow, Advisor: Dr. Yan Ma — *Kean University*
- Building an **AI-assisted interactive Java execution visualizer** for undergraduate CS education; engineered a **trace-ingestion pipeline** parsing **JDI/java_jail** execution traces into a live **stack and heap renderer** with reference arrows; integrating **Google Gemini** for trace-aligned explanations; conducting an **IRB-approved pilot study** measuring learning outcomes and cognitive load vs. static materials.

NSF IUSE Research Scholar | *Jobs vs. Wages* — May. 2026 – Present
Undergraduate Researcher, Advisor: Prof. Pauline Wang — *Kean University*
- Built a **3-tier interactive data visualization platform** (**PHP**, **MySQL**, **JavaScript**, **Google Charts**) with normalized **data marts** aggregating 33,000 ZIP-level records into 3,203 county rows; engineered dynamic charts computing **Pearson Correlation Coefficients** and **linear regression trendlines** across **4 variable pairs**, identifying college education as the strongest household income predictor ($r = 0.7872$). Accepted to **GMiS Research Posters Competition**.

CRA UR2PhD Program — Feb. 2026 – Apr. 2026
Undergraduate Research Trainee
- Co-authored the "Debate Coach" proposal, designing a 10-week framework to **benchmark fine-tuned LLMs** as conversational debate coaches using **GPT-4-as-a-judge**; discovered fine-tuning yielded a **100% hallucinated source rate**, demonstrating the need for **RAG**; presented findings **in person** at the **2026 UR2PhD Mentoring Workshop & Research Showcase** in New Orleans.

Projects

Adzio | *Next.js, Tailwind, Framer Motion, GSAP* | adzio.io — Feb. 2026 – Aug. 2026
- Rebuilt the **marketing website** for a **digital agency** client, engineering the **navbar**, **hero**, **process section**, and **scroll-triggered animations** with **Framer Motion** and **GSAP** to align the site with the agency's brand and improve visitor conversion.

AIO Training | *Next.js, Tailwind, Supabase, Vercel* | trainingaio.com — June 2026 – July 2026
- Built a **full-stack fitness booking platform** replacing manual scheduling, enabling trainers to publish **real-time availability** and clients to book and **pay for sessions directly** via integrated **payment processing**; architected the **Supabase schema** for user profiles and session management, **shipped within 2 weeks**, retained for ongoing development.

Day's Ice Cream | *Next.js, TypeScript, Tailwind, Supabase, Vercel* | daysicecream.com — Mar. 2026 – May 2026
- Designed and shipped a **production full-stack website** for a local ice cream shop, engineering a **custom admin edit mode** backed by **Supabase** so non-technical owners can update live site content directly from a **relational database** without developer intervention; handles **600+ monthly visitors**.